

8+ yrs Production AI in retail & automotive	\$1B+ Projected sales impact from CV programmes	5 Patents filed, Mercedes-Benz R&D India	NeurIPS 6DoF pose research, 2020	10+ DS/AI engineers led cross-functionally
---	---	--	--	--

PROFILE

Eight years building AI systems that hold up under real conditions — tight latency budgets, sparse labels, unpredictable environments, and stakeholders who need answers rather than experiments. My work sits at the junction of computer vision, deep learning and applied GenAI, with the bulk of it shipped to production: a perception stack inside every MBUX Interior Assistant vehicle, and multi-camera tracking and VLM systems running across Lowe's store network.

EXPERIENCE

Lead Data Scientist — AI/ML and Computer Vision

Jul 2022 — Present

Lowe's Companies Inc. — Fortune 50 retailer, \$90B revenue, 1,700+ stores · Bengaluru, India

- Run the computer vision and GenAI programme across store intelligence, associate tooling, inventory optimisation and checkout automation, managing 10+ DS/AI engineers alongside product, engineering and business leadership.
- Built a real-time Multi-Camera Multi-Target Tracking system for customer-service alerting; the pilot delivered \$1M+ incremental revenue and underpins a \$1B+ projected sales-impact programme.
- Designed semi-supervised and active-learning pipelines for in-store object detection — +15% mAP with a 40% reduction in manual annotation overhead.
- Improved self-checkout recognition with self-supervised contrastive pre-training — 25%+ throughput gain and a measurably lower loss-prevention error rate.
- Delivered a GenAI voice assistant combining RAG, VLM visual grounding and streaming LLM inference — cut product-query resolution time by 60% in user testing.
- Won 1st Prize at Lowe's AI Hackathon 2025 and 3rd Place at the global AI Hackathon 2024 (200+ teams); three proposals selected for the executive AI strategy pipeline.

Senior ML Research Engineer

Dec 2018 — Jun 2022

Mercedes-Benz Research and Development India · Bengaluru, India / Stuttgart, Germany

- Owned the in-cabin perception stack for MBUX Interior Assistant — multi-camera human pose estimation, gesture recognition, occupancy detection and scene understanding on embedded automotive ECUs at <20ms latency.
- Improved PCKh accuracy through Student-Teacher training, variable Gaussian heatmaps and limb prediction; held production KPIs under occlusion, clutter, pets and out-of-distribution scenes that broke earlier models.
- Led face recognition and authentication research using metric learning, unsupervised domain adaptation and synthetic-data transfer — cut real-data requirements by 35%.
- Rebuilt distributed training pipelines and drove INT8 quantisation and FPGA integration — saved hundreds of GPU hours and enabled deployment on low-power AUTOSAR ECUs.
- Published at NeurIPS 2020, filed five patents, and received four company awards including two for patent quality.

EARLIER ROLES

Deep Learning Research Engineer — Silversparro Technologies

Jul — Nov 2018 · Gurgaon

CCTV retail models for detection, tracking and age/gender classification. KMeans anchor initialisation for Faster-RCNN cut convergence 50%, with +6% mAP and +7.3% F1.

Research Intern — KAUST, Computational Bioscience Research Center

May — Jul 2017 · Saudi Arabia

Deep neural models for genetic disease-variant classification under Prof. Robert Hoehndorf — ROC-AUC 0.967, AP 0.964 on genome-sequencing test data.

EDUCATION

Indian Institute of Technology Delhi

2013 — 2018

B.Tech and M.Tech Dual Degree, Mathematics and Computing · New Delhi, India

Master's thesis: *Calculus on Normed Vector Spaces*, supervised by Prof. Dr. Amit Priyadarshi. Head TA for Computer Programming, 60+ students. Coursework in machine learning, linear algebra, probability and stochastic processes, optimisation, image processing, data structures and algorithms.

TECHNICAL SKILLS

DEEP LEARNING	PyTorch, TensorFlow, Keras, Detectron2, TensorRT, ONNX, CUDA, distributed and multi-GPU training
COMPUTER VISION	MTMC tracking, object detection, human pose estimation, gesture and face recognition, 6DoF pose, 3D vision, SLAM, camera calibration, segmentation, OpenCV, DeepStream
GENERATIVE AI	HuggingFace Transformers, LangChain, RAG, GraphRAG, VLMs, agentic AI, multimodal reasoning, visual grounding, LLM fine-tuning
RESEARCH METHODS	Semi-supervised, self-supervised and active learning, domain adaptation, metric learning, synthetic data, contrastive learning, uncertainty-aware evaluation
MLOPS / SYSTEMS	Docker, Kubernetes, CI/CD, Linux, Git, model serving and monitoring, edge AI, INT8 quantisation, FPGA integration, NVIDIA GPU workflows
LANGUAGES	Python (expert), C++, SQL, MATLAB, Lua, Pandas, NumPy, scikit-learn

PUBLICATIONS AND PATENTS

End-to-End Differentiable 6DoF Object Pose Estimation with Local and Global Constraints

DiffCVGP Workshop, NeurIPS 2020 — peer-reviewed publication

PATENTS FILED — MERCEDES-BENZ R&D INDIA, 2020-2022

- Dynamic sound-zone adjustment via 3D head tracking
- Hand cropping for hand-pose recognition
- In-cabin occupancy detection
- In-car entertainment recommendation system
- 6DoF child-seat pose estimation inside a passenger vehicle

AWARDS AND RECOGNITION

Lowe's 2025	1st Prize, AI Hackathon — CV-driven inventory optimisation, planogram compliance, traffic-aware stocking, out-of-stock detection
Lowe's 2024	3rd Place, Techwide AI Global Hackathon (200+ teams) — multimodal GenAI application
Lowe's 2023-25	Three AI proposals selected for the executive AI strategy and innovation pipeline
Mercedes 2022	High Quality Patent Award and Implemented Product Patent Award
Mercedes 2021	Product Innovation Award, Top 5 Products
Mercedes 2020	Silver Award, Best Performing Team Member
India 2011-13	National Merit Award, Government of India — top board performer, Class X and XII

LEADERSHIP AND COMMUNITY

- Own research-to-production delivery end to end — ML design reviews, evaluation plans, model hardening and executive presentations from the same seat.
- Designed and ran enterprise AI workshops for 100+ engineers and business leaders at Lowe's, focused on practical CV and GenAI adoption.
- Mentored junior ML engineers at Mercedes-Benz and Lowe's in experiment design, production hardening, patent writing and research communication.
- Core Member and Technical Executive, Sparks Forum educational NGO, Assam (2014-2018). Executive, Entrepreneurship Development Cell, IIT Delhi (2014-2015).

RESEARCH AND PRODUCT THEMES

PERCEPTION

Annotation-efficient CV, MTMC tracking, pose and gesture estimation, real-time video, edge AI under hard latency and power constraints.

MULTIMODAL AI

RAG pipelines, VLMs, visual grounding, agentic workflows, multimodal retrieval, GenAI for operational decision support.

SYSTEMS AND SCALE

Research-to-production ownership, model quantisation, GPU pipeline optimisation, hardware-aware deployment, cross-functional execution.